I think we can make it even better by combining both into a roadmap that follows the natural progression of an ML engineer—from programming fundamentals to production-ready AI systems. This is the roadmap I would follow if I were starting from scratch in **2026**.

🚀 Complete Core ML Engineer + GenAI Roadmap (12 Months)

```
Programming Foundations
        ↓
Python
        ↓
Git + GitHub + Linux
        ↓
Mathematics
        ↓
NumPy + Pandas + SQL
        ↓
Data Analysis (EDA)
        ↓
Classical Machine Learning
        ↓
Feature Engineering
        ↓
Deep Learning (PyTorch)
        ↓
Computer Vision
        ↓
Natural Language Processing (NLP)
        ↓
Transformers
        ↓
Large Language Models (LLMs)
        ↓
Generative AI (RAG, Agents)
        ↓
Deployment (FastAPI, Docker)
        ↓
MLOps
        ↓
Cloud (AWS)
        ↓
Interview Preparation
```

🗓️ Month 1 — Programming Foundations

Python

- Variables & Data Types
- Loops
- Functions

- OOP
- File Handling
- Exception Handling
- Modules & Packages
- Virtual Environments (`venv`)
- pip

Git & GitHub

- Git basics
- Branching
- Merging
- Pull Requests
- GitHub workflows

Linux

- File system
- Terminal commands
- Permissions
- Package management

Practice

- 100–150 Python coding problems

Mini Project

- Student Management System (CLI)

📅 Month 2 — Mathematics + Python Libraries

Mathematics

Linear Algebra

- Vectors
- Matrices
- Dot Product
- Matrix Multiplication
- Eigenvalues (basic)

Statistics

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation
- Covariance

Probability

- Conditional Probability
- Bayes Theorem
- Normal Distribution

Calculus

- Derivatives
- Partial Derivatives
- Gradient
- Chain Rule

NumPy

- Arrays
- Broadcasting
- Vectorization
- Matrix Operations

Pandas

- DataFrames
- Filtering
- GroupBy
- Merge
- Missing Values

SQL

- SELECT
- JOIN
- GROUP BY
- HAVING
- Window Functions

📅 Month 3 — Data Analysis & Visualization

EDA

- Missing values
- Outliers
- Data Cleaning
- Feature Engineering
- Correlation Analysis

Visualization

- Matplotlib
- Histograms
- Scatter Plots
- Boxplots
- Heatmaps

Scikit-learn Preprocessing

- Scaling
- Encoding
- Train/Test Split

Projects

- Titanic Survival Prediction
- Netflix Data Analysis
- House Price EDA

📅 Months 4–5 — Classical Machine Learning

Regression

- Linear Regression
- Polynomial Regression
- Ridge
- Lasso
- ElasticNet

Classification

- Logistic Regression
- Decision Tree
- Random Forest
- Naive Bayes
- SVM
- KNN

Clustering

- K-Means
- DBSCAN
- Hierarchical Clustering

Dimensionality Reduction

- PCA

Model Evaluation

- Accuracy
- Precision
- Recall
- F1
- ROC-AUC
- Cross Validation

Hyperparameter Tuning

- GridSearchCV
- RandomSearchCV

Projects

- Loan Approval Prediction
- Customer Churn Prediction
- Credit Card Fraud Detection
- House Price Prediction

📅 Months 6–7 — Deep Learning (PyTorch)

Neural Networks

- Perceptron
- Activation Functions
- Forward Propagation
- Backpropagation
- Gradient Descent
- Optimizers
- Loss Functions
- Regularization
- Batch Normalization
- Dropout

PyTorch

- Tensor Operations
- Dataset & DataLoader
- Model Training
- Saving & Loading Models

CNN

- Convolution
- Pooling
- Transfer Learning

Projects

- MNIST Digit Classifier
- Fashion-MNIST
- Cat vs Dog Classifier
- Plant Disease Detection

Month 8 — NLP Fundamentals

Traditional NLP

- Text Cleaning
- Tokenization
- Stop Words
- Lemmatization
- TF-IDF
- Bag of Words
- Word2Vec

- GloVe

Sequence Models

- RNN
- LSTM
- GRU (concepts)

Projects

- Spam Detection
- Sentiment Analysis
- News Classification

Months 9–10 — Transformers & LLMs

Transformers

- Attention
- Self-Attention
- Multi-head Attention
- Positional Encoding
- Encoder
- Decoder

Hugging Face

- Transformers
- Datasets
- Tokenizers
- Pipelines

LLM Concepts

- GPT
- BERT
- T5
- Embeddings
- Prompt Engineering

Projects

- Text Summarization
- Question Answering

- Translation
- Named Entity Recognition

📅 Month 11 — Generative AI

RAG

- Embeddings
- Chunking
- Retrieval
- Re-ranking
- Context Injection

Vector Databases

- FAISS
- Chroma

AI Agents

- LangGraph
- LangChain (basics)
- Function Calling
- Tool Calling

Fine-Tuning Concepts

- PEFT
- LoRA
- QLoRA

Projects

- PDF Chatbot
- Resume Analyzer
- AI Tutor
- Company Knowledge Assistant

📅 Month 12 — Deployment, MLOps & Cloud

FastAPI

- REST APIs
- Model Serving

Docker

- Containers
- Docker Compose

Streamlit / Gradio

- ML Dashboard
- AI Demo Apps

MLOps

- MLflow
- DVC
- CI/CD Concepts
- Model Monitoring
- Model Versioning

Cloud (AWS)

- EC2
- S3
- IAM
- Deploying ML Models

Final Capstone Project

Build a production-ready AI application, for example:

- User authentication (optional)
- FastAPI backend
- PyTorch/Hugging Face model
- RAG pipeline
- Vector database
- Dockerized deployment
- Cloud deployment
- GitHub repository with documentation

🛠️ Essential Tech Stack

Category	Technologies
Language	Python
Version Control	Git, GitHub
OS	Linux
Math	Linear Algebra, Statistics, Probability, Calculus
Data	NumPy, Pandas
Database	SQL
Visualization	Matplotlib
ML	Scikit-learn
Deep Learning	PyTorch
Computer Vision	OpenCV (later), TorchVision
NLP	NLTK, spaCy
Transformers	Hugging Face
GenAI	LangGraph, LangChain (basics), FAISS, Chroma
API	FastAPI
Deployment	Docker
Demo UI	Streamlit, Gradio
MLOps	MLflow, DVC
Cloud	AWS

📚 Best Resources

Programming

- **CS50P (Harvard)** → Python
- **Automate the Boring Stuff with Python**

Mathematics

- **3Blue1Brown** (Linear Algebra & Calculus intuition)
- **Khan Academy**

Machine Learning

- **Andrew Ng – Machine Learning Specialization (Coursera)**

Deep Learning

- **DeepLearning.AI – Deep Learning Specialization**

PyTorch

- **Official PyTorch Tutorials**

SQL

- **SQLBolt**
- **Mode SQL Tutorial**

NLP & LLMs

- **Hugging Face Course** (free)

MLOps

- **Made With ML**
- **MLflow Documentation**

🎯 Portfolio Goal (By the End)

By the time you're job hunting, aim to have:

- ☑ 10–12 polished GitHub projects
- ☑ 2–3 deployed ML/GenAI applications
- ☑ Strong Python and SQL skills
- ☑ Solid understanding of ML and Deep Learning fundamentals
- ☑ Hands-on experience with LLMs, RAG, and Hugging Face
- ☑ FastAPI + Docker deployment experience
- ☑ Basic MLOps and AWS knowledge

One suggestion that will make a huge difference

Don't just follow this roadmap passively. Treat each month like a mini-semester:

- **Week 1–2:** Learn the concepts.
- **Week 3:** Solve coding exercises and small practice problems.
- **Week 4:** Build and document a project, push it to GitHub, and write a LinkedIn post explaining what you learned.

That cycle of **Learn → Practice → Build → Share** will help you retain knowledge far better than watching tutorials alone and will give you a portfolio that stands out when applying for ML engineering roles.

Updated 12-Month Plan

Month	ML Roadmap	DSA Roadmap
1	Python, Git, Linux	Arrays, Strings
2	Math, NumPy, Pandas, SQL	Linked List, Stack, Queue
3	EDA, Visualization	Binary Search, Sorting, Recursion
4	Regression, Classification	Trees
5	Advanced ML + Projects	BST, Heap
6	Neural Networks, PyTorch	Graphs
7	CNNs	Dynamic Programming
8	NLP	Greedy, Sliding Window
9	Transformers	Mixed Practice
10	LLMs	Mixed Practice
11	GenAI (RAG, Agents)	Mock Interviews
12	Deployment, Docker, MLOps, AWS	Interview Preparation